

AdS/CFT

Geometry, the dictionary, and controlled calculations

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The useful content of a duality is a map between questions. This note develops that map from the geometry of anti-de Sitter space to boundary correlation functions, thermodynamics, transport, and entanglement. The emphasis is on calculations whose inputs and approximations can be stated explicitly.

The intended background is quantum field theory, elementary differential geometry, and the variational principle. The main example is the duality between four-dimensional $\mathcal{N} = 4$ super Yang–Mills theory and type IIB string theory on $\text{AdS}_5 \times S^5$. Other examples are identified separately.

Read Sections 1–7 for the basic dictionary, 8–12 for scalar correlators and renormalization, and the remaining sections for applications and worked checks. Standard notation follows the cited literature; no auxiliary abbreviations for combinations of physical parameters are needed.

These are pedagogical study notes, with worked calculations and reading references. They summarize established results and indicate the assumptions behind semiclassical or conjectural statements.

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1 What the correspondence asserts

A conformal field theory has local observables but no dynamical spacetime metric. A quantum theory of gravity has a fluctuating geometry and no gauge-invariant local bulk observable at a coordinate point without further specification. AdS/CFT proposes that these very different descriptions can define the same quantum system. Boundary correlation functions supply operationally well-defined observables for the bulk theory.

The canonical example relates $SU(N)$ $\mathcal{N} = 4$ super Yang–Mills theory to type IIB string theory with N units of five-form flux. The decoupled center-of-mass $U(1)$ is not part of the interacting $SU(N)$ theory. The statement is much stronger than a match between a few classical solutions: at finite parameters it is intended to include string excitations and quantum gravity.

Classical Einstein gravity is only a special limit. Large N suppresses quantum loops; large 't Hooft coupling suppresses string-scale curvature corrections. These are different approximations, and taking one does not automatically justify the other. The duality itself has extensive evidence but is not a general theorem derived for all observables and all parameters.

A practical calculation should therefore begin with three questions. Which boundary theory and state are specified? Which bulk fields are retained? Which errors are controlled by inverse powers of N or the coupling? Those questions prevent a useful gravitational model from being mistaken for an exact description of any strongly coupled material.

Reading connection: The proposal originates in Maldacena [1]; the field-theory and string-theory context is developed in [4].

2 Conformal symmetry and radial evolution

In a Euclidean conformal field theory, a scalar primary of dimension Δ transforms under a dilation as

$$\mathcal{O}(x) \longmapsto a^{-\Delta} \mathcal{O}(x/a). \tag{2.1}$$

Translation and rotation invariance constrain a scalar two-point function to depend on $|x - y|$. Scale invariance then fixes it, at separated points, to

$$\langle \mathcal{O}(x)\mathcal{O}(y) \rangle = \frac{C_{\mathcal{O}}}{|x - y|^{2\Delta}}. \quad (2.2)$$

The coefficient depends on operator normalization. Scale invariance alone does not determine interactions or higher-point functions; cross-ratios remain in four-point functions.

Write flat space in polar coordinates and put $r = e^\tau$:

$$dr^2 + r^2 d\Omega_{d-1}^2 = e^{2\tau} (d\tau^2 + d\Omega_{d-1}^2). \quad (2.3)$$

Up to a Weyl transformation, radial evolution becomes time evolution on a cylinder. The dilation operator becomes the cylinder Hamiltonian, with energies measured relative to the appropriate vacuum energy. A local operator inserted at the origin prepares a state on a surrounding sphere.

This observation explains why global AdS energies are related to operator dimensions. It also clarifies why a boundary operator is not simply a bulk particle at a specified radial coordinate. Operators prepare complete states, and the reconstruction of spatially localized bulk excitations usually requires superpositions.

Check. For a free massless scalar in $d > 2$, the propagator behaves as $|x|^{2-d}$. Comparing powers gives $\Delta = (d - 2)/2$, consistent with the kinetic term being dimensionless.

3 The geometry of anti-de Sitter space

The Poincare patch of Lorentzian AdS_{d+1} has metric

$$ds^2 = \frac{L^2}{z^2} (dz^2 - dt^2 + d\mathbf{x}^2), \quad z > 0. \quad (3.1)$$

The boundary is approached as $z \rightarrow 0$. It is not at finite proper radial distance: integrating Ldz/z diverges logarithmically. The boundary metric is defined only up to a Weyl factor because removing the divergent conformal factor is not unique.

The transformation $(z, t, \mathbf{x}) \rightarrow a(z, t, \mathbf{x})$ leaves this metric invariant. This is the bulk realization of a boundary dilation. Boundary distances small

compared with a fixed reference scale correspond to small z ; larger z probes coarser boundary structure. This motivates the radial interpretation as an energy scale, without making the radial coordinate literally equal to an RG cutoff in every setup.

The curvature tensors satisfy

$$R_{ab} = -\frac{d}{L^2}g_{ab}, \quad R = -\frac{d(d+1)}{L^2}. \quad (3.2)$$

For an Einstein action $R - 2\Lambda$, the corresponding cosmological constant is $\Lambda = -d(d-1)/(2L^2)$.

Global AdS instead has a cylinder boundary. Its time coordinate should be unwrapped to avoid closed timelike curves. The global and Poincare descriptions have different coordinate coverage and natural Hamiltonians; their vacuum relations must be handled with the same care as different coordinate systems in ordinary field theory.

Check. The conformal boundary has d spacetime dimensions, whereas the bulk has $d+1$. In formulas below, d always counts the boundary dimensions.

4 D3 branes and decoupling

A stack of D3 branes admits two low-energy descriptions. Open strings ending on the branes give a four-dimensional gauge theory. The near-horizon region of the brane geometry gives a closed-string description. The decoupling argument isolates sectors that cease exchanging finite low-energy excitations with asymptotically flat space.

In the conventional normalization used here,

$$g_{\text{YM}}^2 = 4\pi g_s, \quad L^4 = 4\pi g_s N \alpha'^2, \quad \lambda = g_{\text{YM}}^2 N. \quad (4.1)$$

Consequently,

$$\frac{L^4}{\alpha'^2} = \lambda, \quad g_s = \frac{\lambda}{4\pi N}. \quad (4.2)$$

The numerical factor in the first relation depends on gauge-generator and action conventions; all three relations must be used consistently.

Small curvature in string units requires $L^2/\alpha' \gg 1$, hence $\lambda \gg 1$. Weak string coupling additionally follows in the simple regime $N \gg \lambda$. The

effective gravitational loop parameter in AdS units is proportional to $1/N^2$, so the organization of string perturbation theory and low-energy gravitational perturbation theory should not be conflated.

A string oscillator has mass of order $1/\sqrt{\alpha'}$, so its AdS energy is of order $\lambda^{1/4}$. Heavy string states can be omitted from low-energy supergravity calculations, while their boundary dual operators acquire large dimensions. Protected operators can remain light even at strong coupling.

Reading connection: The D-brane origin is explained in [1,4,5]. The argument identifies matching decoupled descriptions; it is not a perturbative derivation of every finite- N identity.

5 Large N and the bulk expansion

Double-line notation organizes gauge-theory diagrams by the topology of an associated surface. At fixed 't Hooft coupling, a connected vacuum diagram of genus g scales as N^{2-2g} . Boundaries introduced by single-trace insertions modify the power in a normalization-dependent way.

For normalized single-trace operators chosen to have two-point functions of order one, the connected n -point function typically behaves as

$$\langle \mathcal{O}_1 \cdots \mathcal{O}_n \rangle_{\text{conn}} \sim N^{2-n}. \quad (5.1)$$

This is the scaling of weakly interacting bulk particles. Large- N factorization means that disconnected pieces dominate many correlators. It does not mean the original gauge fields interact weakly.

The Einstein action has overall scale

$$I \sim \frac{L^{d-1}}{G_{d+1}} \times \text{dimensionless functional}. \quad (5.2)$$

In the canonical adjoint example, $L^{d-1}/G_{d+1} \sim N^2$. This accounts for the extensive number of boundary degrees of freedom seen in the stress-tensor two-point function and thermal entropy.

A sparse spectrum of low-dimension single-trace operators is an additional diagnostic of a local bulk effective theory with a high cutoff. Large central charge alone is insufficient: there can be too many light higher-spin states for an Einstein approximation. Thus a claim of “holography” must identify

whether it means a full duality, a large- N expansion, or specifically a local low-curvature gravitational description.

Check. Changing the normalization of $\mathcal{O}$ changes every explicit power of N in its correlators. Normalize the two-point function before comparing a diagram with a bulk vertex.

6 The source dictionary

The Euclidean generating functional with source J is

$$Z_{\text{CFT}}[J] = \left\langle \exp\left(\int d^d x J \mathcal{O}\right) \right\rangle. \quad (6.1)$$

In the semiclassical bulk description,

$$Z_{\text{CFT}}[J] \simeq \exp[-I_{\text{ren}}[\phi_{(0)} = J]]. \quad (6.2)$$

The boundary condition, bulk regularity, and chosen saddle are all part of this prescription. Differentiating the logarithm produces connected correlation functions. With the displayed source-sign convention,

$$\langle \mathcal{O} \rangle_J = -\frac{\delta I_{\text{ren}}}{\delta J}. \quad (6.3)$$

Other references put a minus sign in the source coupling, reversing this last convention.

The boundary value of a bulk gauge field sources a conserved current. The boundary metric sources the stress tensor. Bulk gauge redundancy corresponds to boundary Ward identities, not to a global symmetry acting on physical bulk states in the same way as a boundary global symmetry.

The formula is simplest in Euclidean signature, where regularity often selects a unique linear solution. In Lorentzian signature one must specify the state and the real-time contour; a boundary source alone does not choose a Green function.

A useful procedure is to write the regulated action, add counterterms, impose the interior condition, and only then differentiate. Differentiating an unrenormalized boundary term can yield divergent expressions and obscure local contact terms.

Reading connection: The boundary generating-functional prescription was developed in [2,3].

7 Scalar falloffs and operator dimensions

For a scalar with Euclidean action proportional to

$$\frac{1}{2} \int d^{d+1}x \sqrt{g} [(\nabla\phi)^2 + m^2\phi^2], \quad (7.1)$$

the equation of motion is $(\nabla^2 - m^2)\phi = 0$. Near the Poincare boundary, neglecting boundary derivatives gives

$$z^2 \partial_z^2 \phi - (d-1)z \partial_z \phi - m^2 L^2 \phi = 0. \quad (7.2)$$

Substituting a power law gives the indicial relation

$$m^2 L^2 = \Delta(\Delta - d). \quad (7.3)$$

For standard quantization, the larger root is used:

$$\Delta = \frac{d}{2} + \sqrt{\frac{d^2}{4} + m^2 L^2}, \quad \phi \sim z^{d-\Delta} \phi_{(0)} + z^\Delta \phi_{(2\Delta-d)}. \quad (7.4)$$

The first coefficient is the source; the second carries state-dependent response information. Derivative terms and logarithms can appear between these orders, so this is a leading schematic expansion.

A negative bulk mass squared need not be unstable. Reality of the exponents requires the Breitenlohner–Freedman bound $m^2 L^2 \geq -d^2/4$, with appropriate boundary conditions. At equality the roots coincide and logarithmic behavior must be treated separately.

Check. In AdS_5 , $m^2 L^2 = -3$ gives roots 1 and 3. Standard quantization has $\Delta = 3$. A massless scalar instead has $\Delta = 4$, not $\Delta = 0$ in standard quantization.

8 A regular Euclidean solution

Fourier transform along the boundary. For nonzero Euclidean momentum magnitude k , the radial equation becomes

$$z^2 \phi'' - (d-1)z \phi' - (k^2 z^2 + m^2 L^2) \phi = 0. \quad (8.1)$$

The solution regular as $z \rightarrow \infty$ is proportional to

$$z^{d/2} K_{\Delta-d/2}(kz). \quad (8.2)$$

The modified Bessel function decays at large positive argument. The independent I solution grows and is excluded by the Euclidean interior condition for this vacuum problem.

For noninteger $\Delta - d/2 > 0$, the small-argument expansion gives the response-to-source ratio

$$\frac{\phi_{(2\Delta-d)}(k)}{\phi_{(0)}(k)} = \frac{\Gamma(d/2 - \Delta)}{\Gamma(\Delta - d/2)} \left(\frac{k}{2}\right)^{2\Delta-d}. \quad (8.3)$$

The nonanalytic dependence on k determines the nonlocal part of the two-point function. Analytic polynomials in k^2 correspond to contact terms in position space and can depend on renormalization scheme.

When the Bessel order is an integer, the gamma-function expression should be interpreted by a regulated limit together with counterterms. The finite answer includes logarithms of momentum. Ignoring that limit can make an apparently divergent coefficient look like a physical divergence at separated points.

Check. For a dimension-three scalar in a four-dimensional boundary theory, the renormalized nonlocal momentum dependence is $k^2 \log k^2$, up to normalization and the renormalization scale.

9 Renormalization as a boundary variational problem

Put the boundary temporarily at $z = \epsilon$. Integrating the scalar action by parts on a solution leaves

$$I_{\text{reg}} = \frac{1}{2} \int_{z=\epsilon} d^d x \sqrt{\gamma} \phi n^a \partial_a \phi. \quad (9.1)$$

For the region $z \geq \epsilon$, the outward unit normal points toward decreasing z . Missing this sign is a frequent source of inconsistent one-point functions.

The induced measure scales as $\sqrt{\gamma} \sim L^d \epsilon^{-d}$. Inserting the leading scalar falloff produces powers of ϵ and, in special dimensions, logarithms. Local

covariant counterterms built from the induced field, induced metric, and tangential derivatives cancel these divergences.

The renormalized action is

$$I_{\text{ren}} = \lim_{\epsilon \rightarrow 0} (I_{\text{reg}} + I_{\text{ct}}). \quad (9.2)$$

After counterterms have been included, its variation gives a finite canonical momentum conjugate to the source. Away from exceptional logarithmic cases, the nonlocal response is proportional to $(2\Delta - d)\phi_{(2\Delta-d)}$, multiplied by the bulk kinetic normalization and the source-sign convention.

Finite local counterterms remain possible. They change contact terms but not a separated-point power law. This distinction is particularly important when comparing answers obtained using different cutoffs or integration-by-parts conventions.

Reading connection: The systematic geometric procedure is described in de Haro, Solodukhin, and Skenderis [7].

10 Alternative quantization and deformations

In the open mass window

$$-\frac{d^2}{4} < m^2 L^2 < -\frac{d^2}{4} + 1, \quad (10.1)$$

both scalar falloffs can admit a normalizable interpretation with suitable boundary conditions. One can interchange which coefficient is held fixed. The resulting theories have operator dimensions Δ and $d - \Delta$. Endpoints require extra care and are not included in this statement.

Changing the boundary condition changes the theory, rather than merely selecting a different solution within the same theory. The generating functionals are related by an appropriate Legendre transform. A mixed boundary condition implements, in suitable large- N settings, a deformation involving a product of operators.

For an operator of dimension below $d/2$, a double-trace interaction has dimension below d and is relevant. A flow can connect the alternative and standard quantizations. The precise relation between the coefficients of the two falloffs and the deformation coupling depends on normalization, so a bare equality copied between conventions can be misleading.

The scalar unitarity bound in a unitary d -dimensional CFT is $\Delta \geq (d-2)/2$. At the lower endpoint the relation to free fields and logarithmic subtleties must be examined. This boundary condition perspective makes the BF window less mysterious: it is a compatibility statement about bulk energy and boundary unitarity.

Check. In $d = 3$, $m^2 L^2 = -2$ lies strictly inside the window and gives dimensions 1 and 2. Both are above the scalar unitarity bound $1/2$.

Reading connection: See [6,8] for alternative quantization and multi-trace boundary conditions.

11 Interactions and Witten diagrams

A cubic bulk interaction produces a boundary three-point function by integrating three bulk-to-boundary propagators against a common bulk vertex:

$$\langle \mathcal{O}(x_1) \mathcal{O}(x_2) \mathcal{O}(x_3) \rangle \propto \int d^{d+1}X \sqrt{g} K(X; x_1) K(X; x_2) K(X; x_3). \quad (11.1)$$

The propagator is not a normalizable particle wavefunction. It is the solution produced by a boundary source localized at one point.

For three identical scalar primaries, conformal symmetry fixes the result to

$$\frac{C_{\mathcal{O}\mathcal{O}\mathcal{O}}}{|x_{12}|^\Delta |x_{23}|^\Delta |x_{31}|^\Delta}. \quad (11.2)$$

The bulk coupling fixes the coefficient after operator normalization. At four points, exchange diagrams encode intermediate bulk particles, while contact vertices encode local interactions. Boundary operator-product expansions reorganize the same information into conformal blocks.

A local low-energy bulk interaction does not imply locality of a boundary correlator as a function of all insertion points. The boundary theory is local in its own spacetime, but the map between a bulk process and boundary data is nonlocal.

Perturbative Witten diagrams are most useful when the bulk effective theory has a cutoff well above the process of interest. Heavy exchanges can then be represented by a series of higher-derivative contact interactions. If

many higher-spin particles remain light, this expansion generally fails to reduce to a short Einstein-like action.

Check. A three-point function has no conformal cross-ratio. Four generic points do. Therefore conformal invariance alone cannot determine a general four-point correlator.

12 Thermal states and black branes

A translationally invariant thermal state is represented, in an Einstein-gravity model, by a planar AdS black brane:

$$ds^2 = \frac{L^2}{z^2} \left[- \left(1 - \frac{z^d}{z_h^d} \right) dt^2 + \frac{dz^2}{1 - z^d/z_h^d} + d\mathbf{x}^2 \right]. \quad (12.1)$$

Smoothness of the Euclidean geometry at $z = z_h$ fixes

$$T = \frac{d}{4\pi z_h}, \quad s = \frac{1}{4G_{d+1}} \left(\frac{L}{z_h} \right)^{d-1}. \quad (12.2)$$

The entropy is horizon area per boundary spatial volume. Its temperature dependence, $s \propto T^{d-1}$, agrees with conformal scaling.

For the canonical AdS_5 example, dimensional reduction gives $G_5 = \pi L^3/(2N^2)$ at leading order. Substitution yields

$$s = \frac{\pi^2}{2} N^2 T^3. \quad (12.3)$$

At zero coupling the leading entropy density is $2\pi^2 N^2 T^3/3$. Their ratio is $3/4$ in the respective limits. This does not mean the full entropy is coupling independent; the comparison demonstrates an order-one strong-coupling effect.

The planar geometry has no Hawking–Page transition against thermal AdS at a finite temperature on an infinite flat boundary in this conformal setting. A spherical boundary introduces a length scale and a different competition between saddles.

Check. Eliminate z_h before comparing equations of state. A bulk radial coordinate is not a physical boundary temperature until the boundary time normalization is fixed.

13 Real-time response and dissipation

The retarded correlator determines linear response to a weak source. At a black-hole horizon, the retarded prescription imposes an ingoing condition. Near a nonextremal horizon, a mode with time dependence $e^{-i\omega t}$ behaves as

$$\phi \propto (1 - z/z_h)^{-i\omega/(4\pi T)}. \quad (13.1)$$

This sign corresponds to regularity in an ingoing coordinate. An outgoing choice computes a different causal object.

The boundary response follows from the ratio of normalizable to source coefficients after renormalization. Poles occur when an ingoing solution exists with vanishing source. These are quasinormal frequencies. Their negative imaginary parts describe decay in a stable thermal state; an upper-half-plane pole signals an instability.

The shear viscosity is extracted from the stress-tensor retarded function through

$$\eta = -\lim_{\omega \rightarrow 0} \frac{1}{\omega} \text{Im} G_{T_{xy}T_{xy}}^R(\omega, \mathbf{0}). \quad (13.2)$$

For isotropic theories whose shear channel is described by a two-derivative Einstein action, one obtains $\eta/s = 1/(4\pi)$ in units $\hbar = k_B = 1$. Higher-derivative terms and other departures from the assumptions can change the ratio. It is not an unconditional lower bound for all quantum matter.

The low-frequency and zero-momentum limits must be taken in the order appropriate to the observable. Hydrodynamic poles can make apparently similar limit procedures inequivalent.

Reading connection: The real-time prescription and the viscosity result are developed in [9,10].

14 Wilson loops and heavy probes

A fundamental Wilson loop in the boundary theory is related, in the semiclassical string limit, to a worldsheet ending on the loop at the AdS boundary. Its leading expectation value is controlled by the regulated Nambu–Goto area. This probes string tension directly and therefore contains information different from supergravity correlators.

For a long rectangular loop, Euclidean time translation invariance reduces the problem to a static string profile. In Poincare AdS, the energy functional is proportional to

$$\frac{\sqrt{\lambda}}{2\pi} \int dx \frac{\sqrt{1 + (\partial_x z)^2}}{z^2}. \quad (14.1)$$

The absence of explicit x dependence supplies a first integral. The profile reaches a maximum depth and returns to the boundary at the two probe positions. Scaling the separation rescales the entire solution, so conformal invariance implies a potential proportional to the inverse separation.

The bare area diverges near both endpoints. Subtracting the energy of two straight strings removes the heavy-quark masses. The resulting attractive potential scales as $-\sqrt{\lambda}$ divided by separation. This square-root coupling dependence differs from a weak-coupling Coulomb calculation, even though both obey the same distance scaling.

At finite temperature a horizon supplies a new scale and connected world-sheets compete with screened configurations. Details depend on the observable and saddle prescription. The potential of an infinitely heavy external probe should not be identified automatically with a gauge-invariant free energy for a dynamical quark.

Check. A conformal vacuum cannot produce a linear confining potential without an additional scale. Dimensional analysis alone rules out that interpretation of the pure-AdS string solution.

15 Entanglement as a geometric observable

For a static boundary state with a classical Einstein dual, the leading entropy of a region is the area of a homologous minimal bulk surface divided by $4G$. The homology condition is essential: the surface and boundary region must together bound an appropriate bulk region.

For an interval of length ℓ in the vacuum of a two-dimensional CFT, the AdS geodesic is a semicircle in the constant-time Poincare slice. Its regulated length is

$$\text{Length} = 2L \log \frac{\ell}{\epsilon} + O(\epsilon^2/\ell^2). \quad (15.1)$$

Using $c = 3L/(2G_3)$ gives

$$S = \frac{c}{3} \log \frac{\ell}{\epsilon} + \text{constant}. \quad (15.2)$$

The universal logarithmic coefficient matches the field-theory replica calculation. The cutoff-dependent constant is not universal.

For time-dependent states, one uses extremal rather than simply minimal surfaces. At subleading order, bulk entanglement contributes, and the correct surface extremizes generalized entropy. These refinements show why the geometric area formula is a leading approximation rather than the definition of entropy in a quantum field theory.

The geometric dictionary is particularly useful because information-theoretic inequalities constrain admissible bulk geometries. However, deriving an inequality from minimal surfaces proves it only for the class of states and approximation in which that surface prescription is valid.

Reading connection: The leading proposal is [11]. The companion note on entanglement develops the replica derivation, mutual information, and quantum corrections in more detail.

16 Radial flow, locality, and reconstruction

Cutting off the bulk at finite radial position leaves a functional of the fields on that slice. Integrating out the exterior region produces an effective boundary functional. This resembles Wilsonian renormalization, but the relation between radial evolution and a field-theory coarse-graining scheme is not unique.

Bulk equations are second order in the radial direction, so radial data include both fields and conjugate momenta. Field-theory RG equations are often presented as first-order flows of couplings. Reconciling the descriptions requires specifying a state, a variational principle, and which variables represent sources or expectation values.

At leading order, a free bulk field can be reconstructed from a smeared boundary operator in an appropriate causal region. Interactions require multi-trace corrections. Gravity adds constraints: local coordinates are gauge choices, and an observable must be gravitationally dressed. Therefore exact bulk locality is not expected at arbitrary precision in a finite- N theory.

A semiclassical code subspace contains states whose excitations admit a common effective geometric description. Within that restricted setting, information can have more than one boundary representation. This is compatible with quantum error correction because the alternative reconstructions agree on the code subspace, rather than being independent copies of arbitrary quantum information.

Check. A successful reconstruction formula for low-energy states does not establish a state-independent local operator behind every possible black-hole horizon. The domain on which an operator identity holds is part of the claim.

17 Worked calculations: dimensions and thermodynamics

17.1 A scalar near the BF bound

Take $d = 3$ and $m^2 L^2 = -2$. The indicial equation is

$$\Delta^2 - 3\Delta + 2 = 0, \quad (17.1)$$

so the roots are 1 and 2. The bound is $-9/4$, and the alternative-quantization window ends at $-5/4$. Thus this example admits both quantizations. In standard quantization the source scales as z and the response as z^2 ; in the alternative theory their roles are exchanged.

The dimension-one double-trace operator has leading large- N dimension two and is relevant in three dimensions. This dimensional check is necessary before interpreting a mixed boundary condition as a relevant flow.

17.2 The planar equation of state

For a CFT with no chemical potentials, tracelessness of the stress tensor gives energy density $\varepsilon = (d-1)p$. Extensivity gives $\varepsilon + p = Ts$. Hence

$$p = \frac{Ts}{d}, \quad \varepsilon = \frac{d-1}{d}Ts. \quad (17.2)$$

With $s \propto T^{d-1}$, differentiating p reproduces $s = \partial p / \partial T$. The sound speed is

$$c_s^2 = \frac{\partial p}{\partial \varepsilon} = \frac{1}{d-1}. \quad (17.3)$$

This is a conformal thermodynamic result, not an independent prediction unique to gravity. The gravitational calculation fixes the normalization of s and therefore of p and ε .

For $d = 4$, the entropy above gives $p = \pi^2 N^2 T^4 / 8$. All densities refer to the same flat boundary metric and leading large- N , strong-coupling limit.

18 Worked calculation: a geodesic and its cutoff

At constant time in AdS_3 , consider the interval $[-\ell/2, \ell/2]$. A semicircle satisfies

$$x = \frac{\ell}{2} \cos \theta, \quad z = \frac{\ell}{2} \sin \theta. \quad (18.1)$$

Substituting into the line element gives

$$ds = \frac{L}{\sin \theta} d\theta. \quad (18.2)$$

The cutoff is imposed at $z = \epsilon$, so the endpoints occur at angles obeying $\sin \theta = 2\epsilon/\ell$. The length is

$$2L \int_{\arcsin(2\epsilon/\ell)}^{\pi/2} \frac{d\theta}{\sin \theta} = 2L \log \frac{\ell}{\epsilon} + O(\epsilon^2/\ell^2). \quad (18.3)$$

The factor of two comes from the two halves of the semicircle. Missing it gives an incorrect entropy coefficient and is easy to detect by comparison with the CFT central charge.

The exercise illustrates how a physical UV divergence appears as a geometric boundary divergence. The proper length diverges even though the coordinate semicircle is finite. A change of cutoff prescription can shift the additive constant; the derivative with respect to $\log \ell$ remains $c/3$.

For a pure vacuum state on a complete boundary slice, the entropy of the interval equals that of its complement. This equality follows from purity, not from the two regions having similar coordinate lengths on an infinite line. A regulator and the global geometry are needed to formulate the complementary-region calculation consistently.

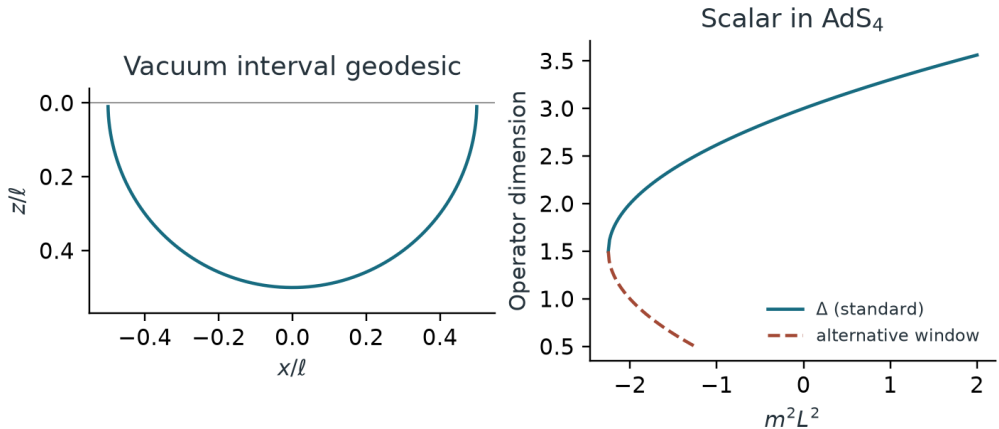


Figure 1: Direct plots of the interval geodesic and scalar mass-dimension relation. The alternative branch is shown only in the open admissible mass window, with endpoints understood as limits. These are analytic illustrations, not numerical gravity solutions.

19 Worked calculation: the black-brane temperature

The Euclidean black-brane time-radial metric is

$$ds_E^2 = \frac{L^2}{z^2} \left[f(z) d\tau^2 + \frac{dz^2}{f(z)} \right], \quad f(z) = 1 - \frac{z^d}{z_h^d}. \quad (19.1)$$

Near the horizon, $f(z) = d(z_h - z)/z_h + \dots$. The conformal factor can be replaced by its finite horizon value at leading order. Set

$$z_h - z = \frac{dz_h}{4L^2} \rho^2. \quad (19.2)$$

Substitution gives

$$ds_E^2 = d\rho^2 + \frac{d^2}{4z_h^2} \rho^2 d\tau^2 + \dots. \quad (19.3)$$

Smooth polar coordinates require $d\tau/(2z_h)$ to have period 2π , so $\beta = 4\pi z_h/d$.

The derivation keeps the conformal factor until proper radial distance has been identified. Dropping it prematurely may accidentally preserve the

final temperature while giving an incorrectly normalized proper distance. The cancellation in the angular coefficient is a useful check rather than permission to ignore metric factors.

The calculation applies to a simple horizon zero. If the zero becomes double, the near-horizon expansion is different and the polar-coordinate argument must be reconsidered. It also assumes the boundary time coordinate appearing in the metric is the one used to define the thermal ensemble.

A practical extension is to repeat the derivation for a spherical AdS black hole. The horizon derivative then contains both a curvature term and a term proportional to the horizon radius. Their competition produces the minimum temperature discussed in the black-hole note.

20 A reading and calculation checklist

Before trusting a holographic result, identify the boundary dimension and signature, the operator normalization, the source sign, and the normalization of boundary time. Distinguish the full string duality from its effective gravitational limit. State whether an observable is evaluated in the vacuum, a thermal ensemble, or a particular excited state.

For scalar calculations, check the BF bound, exceptional logarithmic falloffs, and the interior boundary condition. For metric calculations, include the gravitational boundary terms and counterterms required by the variational principle. For response functions, distinguish Euclidean, time-ordered, and retarded prescriptions.

Three useful exercises extend the worked examples. First, derive the mass-dimension relation directly from the covariant Laplacian, including the determinant of the metric. Second, reproduce the black-brane temperature by expanding the Euclidean metric to polar form at the horizon. Third, compare the scaling of a bulk cubic vertex with the connected three-point function after canonical normalization.

The corresponding checkpoints are $\sqrt{g} \propto z^{-(d+1)}$, Euclidean period $4\pi z_h/d$, and a normalized three-point coefficient of order $1/N$ in the canonical adjoint example. These are independent tests of geometry, regularity, and counting. Passing one does not compensate for failing another.

References and further reading

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